4.2 Analysis and limitations

primo gIveS the intended reply, so as the FAQ, product-search and order-tracking requirements are met for the data supplied and it also guide you about the data required in order to get help from it(primo). The analysis found these limits:

- **Rule order.** "track my delivery" matches the delivery rule first, so tracking is not offered. Testing track before deliver would fix this.
- **Double echo.** The handler writes the customer's message twice, as typed and cleaned (Figure 3). Removing the first write fixes it.
- **Number format.** Prices print from double values (R1099.0); formatting to two decimals is needed.
- **Hard-coded data.** Four laptops, their stock and prices, and four order numbers are fixed in the code, and the stock figures never change. A real shop needs a database.
- **Keyword fragments.** The contains method matches text inside words, so "recall" would trigger the contact reply because it contains "call". Misspellings and several questions in one message are not understood.
- **Missing optional features.** Contact details are given, but there is no live hand-over to a human, and the "help us improve" pop-up does not save feedback, so the bot cannot learn.

5 CONCLUSION AND RECOMMENDATIONS

The group designed, documented and implemented a working Java chat bot called Primo that answers FAQs, searches products and tracks orders, using constructors, modifiers, control statements and a JFrame form as required. Testing showed correct behaviour when asked and answering question about all the frequently asked questions , and it can clearly analyse questions before answering .

We recommend that management (1) correct the rule order, double echo and price format, (2) move products, stock and orders into a database, (3) add synonym lists or a natural language processing library, and (4) store feedback and add live hand-over to staff. With these changes the bot could be piloted with real customers.